

Hallucination Detection Report

Generated: 2026-04-14 10:04 · Extracted triples: 5 · Verdicts: 5 · Inconsistencies: 0 · Symbolic violations: 2

Executive Summary

4 Correct	1 Fabricated Fact	
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Extracted Triples

All verifiable triples extracted from the LLM response before any canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
400	HAS_DOWN_PAYMENT	19.4%	Yes	ENTITY_ATTRIBUTE	Indicates the down payment percentage explicitly associated with the specified credit score.
500	HAS_DOWN_PAYMENT	15.4%	Yes	ENTITY_ATTRIBUTE	Indicates the down payment percentage explicitly associated with the specified credit score.
433	HAS_ESTIMATED_DOWN_PAYMENT	18.4%	Yes	FACTUAL_CLAIM	Indicates the estimated down payment percentage for the specified credit score, derived from interpolation or averaging of provided data.
430	HAS_DOWN_PAYMENT	Not explicitly provided in the dataset	Yes	ENTITY_ATTRIBUTE	Indicates the absence of an explicit down payment percentage for the specified credit score in the dataset.
435	HAS_DOWN_PAYMENT	Not explicitly provided in the dataset	Yes	ENTITY_ATTRIBUTE	Indicates the absence of an explicit down payment percentage for the specified credit score in the dataset.

Canonicalized Triples

Triples after predicate normalization against the database schema. Shows the original predicate, the matched canonical predicate, and match confidence.

Subject	Original Pred.	Object	Canonical Pred.	Confidence
400	HAS_DOWN_PAYMENT	19.4%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH
500	HAS_DOWN_PAYMENT	15.4%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	HIGH
433	HAS_ESTIMATED_DOWN_PAYMENT	18.4%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM
430	HAS_DOWN_PAYMENT	Not explicitly provided in the dataset	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM
435	HAS_DOWN_PAYMENT	Not explicitly provided in the dataset	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM

Logical Inconsistency Report

Contradictions detected between triples in the LLM response — independent of the database. These are self-contradictions within the answer itself.

No logical inconsistencies detected.

Symbolic Rule Violations

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

Constraint ID	Rule Type	Severity	Subject	Predicate	Object	Explanation
c_down_payment_range	RANGE		400	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	19.4%	Down payment the object must be between 0% and 50%. The extracted numeric value 19.4 falls within the valid range, but this triple is being checked for its subject's credit score range, which is not applicable here. However, the object '19.4%' is valid for the down payment constraint.
c3_credit_score_range	RANGE		500	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	15.4%	Credit score the subject must be between 300 and 850 inclusive. The subject '500' falls within the valid range, but this constraint is not relevant for the down payment predicate. The object '15.4%' is valid for the down payment constraint.

Detected Hallucinations

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation / Evidence
400	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	19.4%	CORRECT	STRICT		██████ ████ 1.00	The subject, predicate, and object (with percentage symbol normalized) match the DB candidate exactly. Evidence: subject: 400, predicate: DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION, object: 19.4
500	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	15.4%	CORRECT	STRICT		██████ ████ 1.00	The subject, predicate, and object (with percentage symbol normalized) match the DB candidate exactly. Evidence: subject: 500, predicate: DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION, object: 15.4
433	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	18.4% (estimated)	FABRICATED_FACT	NONE	MULTIPLE	██████ ████ 0.00	No DB evidence exists for the subject or object, and the object makes a concrete verifiable claim (18.4% estimated).
430	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Not explicitly provided in the dataset	CORRECT	SEMANTIC		██████ ████ 0.60	The object semantically conveys the absence of the requested information, which is consistent with the lack of the predicate in DB candidates. Evidence: subject: 430, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 430, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 16.54
435	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Not explicitly provided in the dataset (assumed)	CORRECT	SEMANTIC		██████ ████ 0.60	The object semantically conveys the absence of the requested information, which is consistent with the lack of the predicate in DB candidates. Evidence: subject: 435, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 435, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 16.77

Triple Audit Trail (Original → Canonical → Verdict)

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict (e.g. dropped due to output truncation).

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
400	HAS_DOWN_PAYM ENT	19.4%	HIGH	DOWN_PAYMENT_R EQUIRED_AS_UPFR ONT_BORROWER_C ASH_CONTRIBUTION	19.4%	CORRECT
500	HAS_DOWN_PAYM ENT	15.4%	HIGH	DOWN_PAYMENT_R EQUIRED_AS_UPFR ONT_BORROWER_C ASH_CONTRIBUTION	15.4%	CORRECT
433	HAS_ESTIMATED_D OWN_PAYMENT	18.4%	MEDIUM	DOWN_PAYMENT_R EQUIRED_AS_UPFR ONT_BORROWER_C ASH_CONTRIBUTION	18.4% (estimated)	FABRICATED_F ACT
430	HAS_DOWN_PAYM ENT	Not explicitly provided in the dataset	MEDIUM	DOWN_PAYMENT_R EQUIRED_AS_UPFR ONT_BORROWER_C ASH_CONTRIBUTION	Not explicitly provided in the dataset	CORRECT
435	HAS_DOWN_PAYM ENT	Not explicitly provided in the dataset	MEDIUM	DOWN_PAYMENT_R EQUIRED_AS_UPFR ONT_BORROWER_C ASH_CONTRIBUTION	Not explicitly provided in the dataset (assumed)	CORRECT

Verdict Legend

Verdict	Description
CORRECT	All three components (subject, predicate, object) verified against DB.
RULE_VIOLATION	One or more components contradict DB evidence.
MULTI_RULE_VIOLATION	Violation spans multiple DB triples simultaneously.
UNVERIFIABLE	No usable DB evidence found for any component.
PREDICATE_MISMATCH	Subject real, but stated predicate is not a valid relation for it.
FABRICATED_FACT	Triple asserts a relationship that does not exist anywhere in DB.